

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Owned backend reliability and quality: added integration tests for regression protection, monitored production logs and debugged tenant-specific incidents, hardened auth and rate limiting, and remediated **20+ VAPT findings**.
- Architected a full-stack internal tooling and ML data-annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Served as primary technical POC, wrote knowledge-transfer documentation onboarding a second developer to on-call readiness in **~6 weeks**, and piloted a COO-sponsored GitHub Copilot experiment for AI-assisted development.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ user-facing RESTful APIs** (Node.js, Fastify, PostgreSQL) with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM feature (**4-bit quantized Llama-7B via llama.cpp in C++**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with **git-webhook auto-deploy** (GitHub webhook to a droplet on push), cron automation, and Stripe + manual-cash payments; multi-branch from day one, extended to a 2nd branch with **zero schema migration**; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI service exposing user-facing APIs for a multi-stage discovery pipeline (typed retrieval planning, reranking, deterministic extraction with LLM fallback, cell-level provenance); hardened with **150+ automated tests**, instrumentation, and provider routing/fallback.

Cache- and SIMD-Aware Matrix Multiplication (GitHub) C++, AVX/SIMD, OpenMP, Linux perf

- Optimized C++ matrix multiplication from a naive baseline through loop reordering, cache-aware tiling, manual **AVX/SIMD** vectorization, register-aware micro-kernels, and **OpenMP** multithreading, measuring and validating L1/L3/TLB cache misses with **perf** and inspecting the generated assembly.

AutoEval – Retrieval Evaluation Improvement Agent (GitHub) Python, IR evaluation, Git automation

- Built an automated evaluation-tooling agent that proposes and validates test cases, scores them against multiple reference systems, and commits only improvements to **Git** with report cards and anti-Goodharting guardrails.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: C++, Python, JavaScript, TypeScript, SQL, Java, Bash, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: Node.js, Express, FastAPI, React, Docker, Git, GitHub Actions, REST APIs

Testing / Automation: pytest, integration testing, LambdaTest, CI/CD, Git webhooks, perf profiling, debugging

ML & AI: PyTorch, Hugging Face Transformers, LLM integration, RAG, NumPy, evaluation

Cloud / Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.